Does Adaptive Weighting Improve Equity Market Stress Measurement? Chronological Robustness Evidence from a GARCH–HMM–EVT Composite Index

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Abstract

This paper investigates whether conditional volatility, filtered market-regime probabilities and extreme-loss severity benefit from regime-conditioned entropy weighting in a composite equity-market stress indicator. The Adaptive Composite Market Stress Index (A-CMSI) combines ARIMA–conditional-variance models, forward-filtered Gaussian hidden Markov probabilities and rolling peaks-over-threshold expected shortfall for AAPL, MSFT and SPY. Fresh estimation reproduces the previously reported five-state results on 2,765 adjusted-price observations from 2015–2025 and 1,810 scored 30-day forecasts: Adaptive RMSE is 0.10532, compared with 0.10457 for Equal and 0.10411 for NoHMM. This revision executes complete three-, four-, five- and six-state paths, independently removes panic probability and regime multipliers, evaluates direct and deficit entropy weights, and tests nine EVT window/threshold configurations. The controlled Adaptive specification has RMSE 0.10532 on 1810 matched dates; the lowest point-estimate RMSE among the displayed composite/baseline variants is obtained by NoHMM. Five EVT configurations cannot meet the fixed minimum-exceedance rule and become entirely empirical fallback estimates. The study separates after-close detection from future-event prediction, reports dependence-aware uncertainty and multiplicity-adjusted loss comparisons, and retains unfavorable baselines. Adaptive weighting has no demonstrated predictive advantage over simpler weighting baselines in this sample. The evidence supports a transparent research framework, not a deployment-ready systemic-risk indicator.

Keywords: equity-market stress; composite indicator; GARCH; hidden Markov model; extreme value theory; entropy weighting; chronological validation.
JEL: C58, G01, C53, G17.

1 Introduction

Financial stress is multidimensional. Conditional dispersion, persistent market regimes and the severity of extreme losses need not convey identical information. A composite index can summarize these signals, but a plausible economic narrative is not evidence of improved forecasting. More

elaborate aggregation can add estimation noise, obscure redundant components, or concentrate weight on signals unrelated to the chosen loss. A stress index should therefore be evaluated against simple alternatives without a presumption that adaptive construction must win.

Financial stress composites have substantial precedent. CISS combines market-specific indicators while accounting for cross-market dependence [Holló et al., 2012]. FIMSIS examines stress aggregation in a smaller financial system [Drenkovska and Lenarčič, 2025]. A-CMSI has a narrower scope: two large technology stocks and a U.S. equity ETF containing both stocks. It is an equity stress-measurement experiment, not a replacement for multi-market financial-system indicators. The contribution investigated here is the explicit combination of volatility, filtered panic probability and tail severity with trailing entropy and soft regime factors.

This revision asks three empirical questions. Does adding a sixth HMM state improve generalization despite lower initial-training BIC? Does the panic probability contribute information beyond V/T, and are any losses associated instead with the regime multipliers? Are index conclusions stable to feasible POT thresholds and windows, and to direct versus deficit entropy? All requested specifications are executed under a fixed protocol. Historical outputs are archived, not replaced with a winning selection.

The previous preprint already reported that NoHMM and Equal have lower point-estimate volatility RMSE than Adaptive. The new experiments revisit the same historical period and are therefore retrospective robustness analyses. Their chronological implementation prevents computational look-ahead, but does not convert an already inspected period into an independent confirmatory holdout.

2 Methodological context

ARCH and GARCH model conditional variance [Engle, 1982, Bollerslev, 1986]; EGARCH represents asymmetric responses through log variance [Nelson, 1991]. Latent Markov regimes describe persistent distributional changes [Hamilton, 1989]. POT tail inference is motivated by extreme-value approximations [Pickands, 1975], with financial conditional-tail applications in McNeil and Frey [2000]. Hybrid HMM–EVT risk modeling is also established [Salhi et al., 2016]; A-CMSI does not claim to invent these model families.

Shannon entropy quantifies concentration in a probability distribution [Shannon, 1948]; using it as a component weight is a design choice rather than a theorem about predictive usefulness. The direct and deficit constructions deliberately reward different temporal patterns. Cointegration [Johansen, 1988] and wavelet analysis [Torrence and Compo, 1998] provide descriptive context but are not daily forecasting inputs derived from full-sample smoothing. Forecast comparison follows the squared-loss framework of Diebold and Mariano [1995], with the small-sample horizon adjustment of Harvey et al. [1997] and additional block resampling.

3 Data, chronology and reproducibility

The canonical local input contains 2,765 complete adjusted-price rows from 2 January 2015 through 30 December 2025. For asset a, daily log return is

$$r_{a,t} = \log \Pi_{a,t} - \log \Pi_{a,t-1}, \tag{1}$$

where Π denotes adjusted price, distinct from the panic component P_t. This yields 2,764 return dates. Prices are checked for missing and nonpositive values, sorted unique dates and alignment.

The saved VIX observations start on 4 January 2016; earlier missing VIX values are not backfilled using future data. Legacy saved returns differ very slightly from recomputation, so the operational pipeline always derives returns from the canonical dated price file.

The initial model window contains 756 return observations. Selected mean/variance specifications and HMM parameters are re-estimated every 21 trading days. The next block receives fixed estimated parameters; after-close observations update residuals and filtered states sequentially. The historical initial reconstruction initializes rolling features and is excluded from forecast calibration/scoring. All scalers use training rows only. Conditional-variance and filtering prefix tests verify that extending a fixed-parameter sequence cannot revise its earlier operational values.

The original scored period covers 5 September 2018–14 November 2025. The last 30 signal dates still receive forecasts but cannot be scored without future observations. Longer tail windows change warmup; new comparisons are restricted to identical training support and scoring dates within each family, with a second intersection shared across all families. Tables state their own sample sizes. They must not be ranked across different eligible periods as though the observations were identical.

The audit stores full SHA-256 hashes of every input CSV, source hashes, exact package versions, configuration, seeds, timestamps and logs. Fresh fits reproduce the original tables and predictions. The original production implementation and observed data remain unchanged. The full fresh-fit reproduction used the recorded project environment. A second fully isolated installation from the exact dependency lock passed all 17 tests and reproduced calibrated forecasts exactly, without project dependency-folder or global-package fallback. These two checks are distinguished rather than claiming that every full model experiment was repeated in the isolated environment.

4 Components and index construction

4.1 Conditional volatility

Initial-training BIC selects ARIMA$(p, 0, q)$ with $p, q \in \{0, 1, 2\}$. Innovations feed zero-mean GARCH, GJR-GARCH and EGARCH candidates with normal and Student-t errors. Selected specifications are re-estimated monthly. Returns are scaled to percentage units during variance fitting and exported in decimal-return units. Conditional standard deviation supplies raw volatility $\sigma_{a,t}$.

For a generic component X, let trailing normalization be

$$Z_t(X) = \mathrm{clip}_{[-3,3]}\left[\frac{X_t - \bar{X}_{t,252}}{s_{t,252}(X)}\right], \quad \bar{X}_{t,252} = \frac{1}{252}\sum_{k=0}^{251} X_{t-k}. \tag{2}$$

Sample standard deviation with denominator 251 is used; incomplete windows and zero-variance normalizations remain missing. Today's value is included because signals are after close. Thus $V_{a,t} = Z_t(\sigma_a)$. No full-sample normalization is used.

4.2 Forward-filtered regimes

The multivariate Gaussian HMM observes SPY return, normalized conditional volatility, 21-day realized volatility and skewness, drawdown, volatility of volatility, VIX and mean pairwise return

correlation. For transition matrix A, state density f_s, and filtered probability $\alpha_t(s)$,

$$\alpha_t(s) = \frac{f_s(x_t) \sum_j \alpha_{t-1}(j) A_{js}}{\sum_\ell f_\ell(x_t) \sum_j \alpha_{t-1}(j) A_{j\ell}}. \tag{3}$$

No backward smoothing conditions today's probability on later observations. Initialize the first posterior proportional to start probabilities times the first observation likelihood; the transition recursion applies from the second observation onward. At every refit, training means rank states using VIX, realized volatility, conditional volatility and inverse return. The highest-risk state defines raw panic probability; normalized $P_t = Z_t(\alpha_t(s_{\text{panic}}))$ is common to all asset indices. Numeric state IDs themselves have no persistent economic meaning.

The original model uses five states. The audit executes three to six states at all 96 origins with seeds 42, 43 and 44 and at most 200 EM iterations. Maximum training likelihood chooses the restart. Iteration limits and negative final likelihood increments are reported rather than hidden behind a library convergence flag. Occupancy is computed from filtered responsibilities; state-duration estimates are $1/(1 - A_{ss})$. State stability compares common 126-observation training anchors with permutation- invariant ARI and Hungarian-aligned labels. Symmetric Gaussian KL diagnoses near-redundant emissions.

4.3 Tail severity

Let loss $L_{a,t} = -r_{a,t}$. For trailing window N and threshold quantile q, the POT threshold u_t is estimated only from that window. Positive excesses $Y = L - u_t$ follow the fitted GPD

$$G(y) = 1 - (1 + \xi_t y/\beta_t)^{-1/\xi_t}, \tag{4}$$

on its valid support, with fixed location zero and logarithmic/exponential limits at $\xi = 0$. If $p_{u,t} = n_{u,t}/N$ is the exceedance rate, then

$$\text{VaR}_{\tau,t} = u_t + \frac{\beta_t}{\xi_t} \left[\left(\frac{p_{u,t}}{1-\tau} \right)^{\xi_t} - 1 \right], \tag{5}$$

$$\text{ES}_{\tau,t} = \frac{\text{VaR}_{\tau,t} + \beta_t - \xi_t u_t}{1 - \xi_t}, \quad \xi_t < 1. \tag{6}$$

Here τ is the loss-quantile confidence level, distinct from HMM probabilities $\alpha_t(s)$, and extrapolation requires $1 - \tau \le p_{u,t}$. At $\xi_t = 0$, VaR is $u_t + \beta_t \log[p_{u,t}/(1-\tau)]$. The baseline uses $N = 252$, $q = 0.90$, at least 20 exceedances, and $\tau = 0.99$. When insufficient excesses, numerical failure or infinite fitted ES makes the GPD unusable, the operational raw tail score uses empirical ES99. The failure status and missing parametric quantity are retained. Thus $T_{a,t}$ is a normalized *hybrid* tail-severity estimator, not an always-successful GPD estimate. Shape is never clipped to manufacture a finite mean. The fallback averages observed losses at or above the window's empirical 99th-percentile quantile, without fractional weighting of the boundary observation. $T_{a,t}$ is the raw operative tail score passed through $Z_t(\cdot)$ once.

The sensitivity grid is $N \in \{252, 378, 504\}$ and $q \in \{0.90, 0.95, 0.975\}$, with the original 20-exceedance constraint and 252-day normalization unchanged. Static full-sample fits are labeled descriptive and never select the forecast specification.

4.4 Entropy and adaptive weights

For asset a, define the already normalized vector by $Z_{a,V,t} = V_{a,t}$, $Z_{a,P,t} = P_t$, and $Z_{a,T,t} = T_{a,t}$; no second normalization is applied. For component $i \in \{V, P, T\}$ and trailing entropy window $W = 63$, define probabilities across time:

$$p_{a,i,k,t} = \frac{|Z_{a,i,t-k}|}{\sum_{j=0}^{W-1} |Z_{a,i,t-j}|}, \quad H_{a,i,t} = -\frac{\sum_{k=0}^{W-1} p_{a,i,k,t} \log p_{a,i,k,t}}{\log W}. \tag{7}$$

The convention is $0 \log 0 = 0$. A complete zero-magnitude component window has $H = 0$. Direct entropy weights normalize $H_{a,i,t}$ across components; deficit weights normalize $1 - H_{a,i,t}$. If that cross-component total is zero, base weights are equal; incomplete windows remain missing. Direct entropy rewards sustained temporal activity, while deficit weights emphasize concentrated activity. Neither definition identifies economically useful information without empirical testing.

Let ρ_s be evenly spaced risk ranks from zero to one, assigned from training means, and $S_t = \sum_s \alpha_t(s)\rho_s$. Soft factors interpolate the documented bull/panic endpoints:

$$f_{V,t} = 1.3 - 0.5S_t, \quad f_{P,t} = 0.8 + 0.3S_t, \quad f_{T,t} = 0.9 + 0.5S_t. \tag{8}$$

This is the historical extension to ranked five-state probabilities, not a newly estimated multiplier rule. It does not exactly preserve the neutral vector of the original conceptual three-regime table. For asset-specific entropy base $e_{a,i,t}$, raw weight $\omega^{\text{raw}}_{a,i,t} = e_{a,i,t} f_{i,t}$ receives the exact floor map

$$w_{a,i,t} = m + (1 - Mm)\frac{\omega^{\text{raw}}_{a,i,t}}{\sum_j \omega^{\text{raw}}_{a,j,t}}, \quad m = 0.05, \tag{9}$$

where M is the number of included components and $0 \le m < 1/M$; K denotes the HMM state count. Weights differ by asset, even though panic probability and regime factors are shared. This guarantees nonnegative weights, the minimum and a unit sum simultaneously. The final index is

$$\text{ACMSI}_{a,t} = \sum_i w_{a,i,t} Z_{a,i,t}, \qquad \text{ACMSI}_{\text{portfolio,t}} = \frac{1}{3} \sum_{a \in \{\text{AAPL,MSFT,SPY}\}} \text{ACMSI}_{a,t}\,. \tag{10}$$

An optional fourth component uses normalized mean rolling pairwise correlation with a neutral factor. High correlation alone does not show all assets are falling and is not termed cointegration breakdown.

5 Experimental design and inference

5.1 Controlled comparisons

NoProbability removes P but retains V/T regime factors; Entropy removes factors but retains P; NoHMM removes both. This distinction prevents the no-HMM ablation from merely duplicating entropy-only. Equal, original absolute-loading PCA, NoEVT, NoGARCH and Adaptive4 remain visible. PCA uses earlier observations only. The original PCA method does not impose the minimum floor; PCAFloor is a separately reported constrained alternative. Original entropy exports are pre-floor weights, whereas the entropy index applies the floor. The audit does not silently rewrite the original scores.

ConditionalVT, ConditionalVTP, ConditionalVTStress and ConditionalVTPStress use separate V/T/P/stress regressors in otherwise fixed ridge/logistic calibration. Ridge uses regularization $\alpha_{\text{ridge}} = 1$; logistic regression uses inverse regularization $C = 1$. They are exploratory incremental-information diagnostics, not new composite weights. No hyperparameter is optimized on evaluation outcomes.

5.2 Forecasting versus detection

The future volatility target is the next 30 daily returns' sample standard deviation annualized by $\sqrt{252}$. Standardized ridge calibration with penalty one predicts log volatility; exponentiation produces forecasts. This preserves the original method, without a lognormal mean-bias correction. A training label is admitted only if its entire 30-day future window ends before the next predicted block. At least 126 mature operational observations are needed. Training means and current 30-day realized volatility provide constant and persistence baselines.

A contemporaneous crisis label is one for SPY daily log return below -0.03 or trailing 20-day simple cumulative price return below -0.10. Same-day index threshold scores are *after-close detection*. Separate logistic models with inverse regularization $C = 1$ forecast any crisis label in the next 30 trading days. Precision, recall and F1 use a predeclared probability cutoff of 0.5; ROC AUC is threshold-free. Expanding training prevalence is supplied as a probability baseline. For the TrainMean and RealizedVolBaseline rows, crisis probabilities use this training prevalence; those rows do not represent separate volatility-driven crisis classifiers. The future filtered-regime transition target remains a model-derived proxy and does not substitute for observed crisis labels.

5.3 Uncertainty and testing

For forecasts A and B, squared-loss difference $d_t = (y_t - \hat{y}_{A,t})^2 - (y_t - \hat{y}_{B,t})^2$ enters a DM statistic with Bartlett HAC long-run variance through lag 29. The HLN correction at horizon $h = 30$ multiplies the statistic by

$$\sqrt{\frac{n + 1 - 2h + h(h-1)/n}{n}}. \tag{11}$$

Following the dependent-data block-bootstrap principle [Künsch, 1989], the circular moving-block implementation uses 30-day blocks and 1,000 replicates, resampling outcomes and predictions together. The 60-day sensitivity changes block length and HAC bandwidth to 59. Holm corrections [Holm, 1979] are supplied within each stated squared-loss family and across the full comparison set. Bootstrap intervals for other metrics are marginal descriptive intervals, not simultaneous bands. AUC draws without both classes are omitted and counted. Nonsignificance is not evidence of equivalence.

6 Results

6.1 Fresh reproduction

Fresh five-state estimation reproduces the original RMSE, MAE, directional accuracy and crisis metrics, together with saved predictions, components and initial-training BIC comparisons. Original and revised outputs are stored separately. The published negative result is retained: Adaptive does not establish a forecasting advantage over Equal or NoHMM in that sample.

Table 1: Original results reproduced from fresh estimation. Annualized volatility uses decimal-return units.

Model	Dates	RMSE	MAE	Future-crisis AUC
NoHMM	1810	0.1041	0.0564	0.6573
Equal	1810	0.1046	0.0568	0.6776
Entropy	1810	0.1048	0.0568	0.6656
Adaptive4	1810	0.1053	0.0576	0.6611
PCA	1810	0.1053	0.0570	0.6843
Adaptive	1810	0.1053	0.0569	0.6583
NoGARCH	1810	0.1058	0.0572	0.6468
NoEVT	1810	0.1068	0.0597	0.5919
RealizedVolBaseline	1810	0.1227	0.0686	–

6.2 State count and generalization

Table 2: State-count robustness on identical chronological scoring dates. The model is the adaptive composite, not the HMM alone.

States	Dates	RMSE	95% block CI	MAE	AUC	Brier
HMM3	1810	0.1067	[0.0577, 0.1563]	0.0590	0.6164	0.1974
HMM4	1810	0.1069	[0.0574, 0.1574]	0.0591	0.6395	0.1929
HMM5	1810	0.1053	[0.0557, 0.1555]	0.0569	0.6583	0.1869
HMM6	1810	0.1050	[0.0559, 0.1544]	0.0570	0.6481	0.1866

HMM6 has the smaller point-estimate volatility RMSE. The five-versus-six comparison does not establish a family-adjusted volatility advantage; the RMSE difference remains uncertain. This is not proof of equivalence. Training BIC alone does not establish out-of-sample overfitting.

Six states increase average training log density by 0.354 per observation, but change held-out conditional log density by -2.090 per day relative to five (three-month-block 95% bootstrap interval [-7.601, 1.375]). This train/test divergence is evidence consistent with overfitting for emission-density estimation; it does not automatically establish a volatility-forecast disadvantage. The interval includes zero, so a density disadvantage is not statistically established. The density interval is exploratory and not a multiplicity-adjusted confirmatory test.

Emission-density diagnostics cover all 2,008 post-initial-training return days, whereas the calibrated volatility forecasts cover 1,810 scoring dates. The three-month-block density comparison was added after inspecting the train/test density pattern and is explicitly exploratory.

The density result and index-forecast result address different targets. More complex Gaussian emissions can improve training fit while assigning poor conditional density to later observations; a nonlinear filtered-probability index need not exhibit the same ranking under volatility RMSE. Raw

Table 3: HMM diagnostics; K is state count. Rare-state fraction is across state/refit records, not days. Rare means expected occupancy below max(20,1 percent of training rows). Duration is $1/(1 - A_{ss})$; duration and ARI columns report medians.

K	Initial BIC	OOS log density/day	Rare-state fraction	Median ARI	Median duration	Nonmonotone fits
3	6791.0640	-11.1151	0.0000	0.9633	43.0870	0
4	5917.5216	-11.6698	0.0000	0.5750	32.1628	0
5	5568.1103	-12.8373	0.0083	0.5551	26.4765	1
6	5532.2710	-14.9269	0.0122	0.4340	22.9985	1

state IDs cannot be compared across fits without alignment. Rare occupancy, refit instability and close emission distributions are diagnostics rather than decisive individual tests of overfitting. The historical default is retained for continuity, not selected anew from these inspected outcomes.

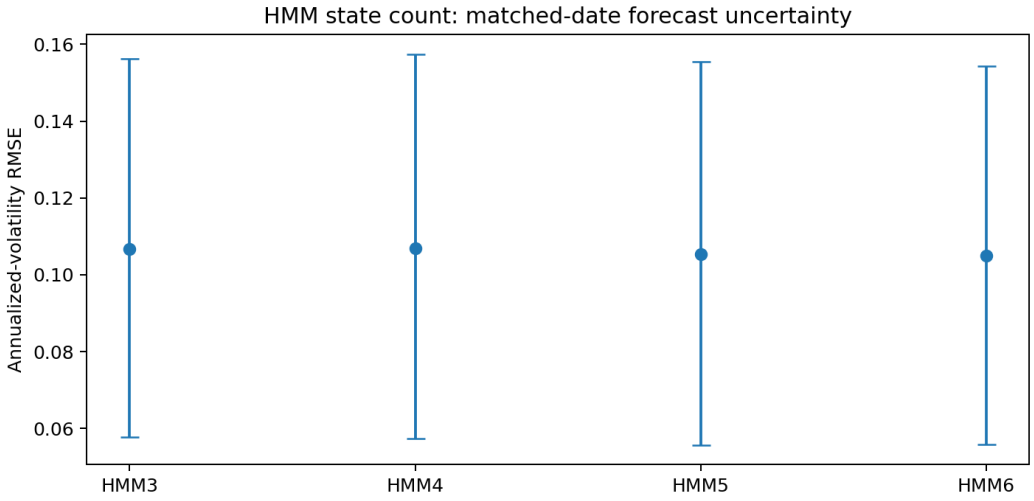


Figure 1: Matched state-count RMSE intervals. Marginal interval overlap is not itself a test of a paired difference.

6.3 Incremental HMM contribution and weighting

Removing P alone gives RMSE 0.10519; removing multipliers alone gives 0.10478; removing both gives 0.10411, compared with 0.10532 for Adaptive. The paired RMSE intervals for these three Adaptive contrasts include zero. This pattern is consistent with redundant/noisy contributions under this fixed calibration, but it does not identify a unique mechanism or prove that HMM information is useless. The lowest composite point estimate is NoHMM, 0.10411. Full paired effect sizes and adjusted loss-test p-values are reported rather than inferring significance from point rankings. Removing P and removing regime factors are different interventions. Improvements must be read with paired uncertainty, rather than assigned to an assumed mechanism based on a single ranking. The conditional regressions ask whether P/stress has incremental information beyond V/T under the same fixed calibration. They do not establish structural causality or prove absence of information when an interval contains zero. PCAFloor explicitly tests a constraint correction while preserving the original unconstrained PCA baseline.

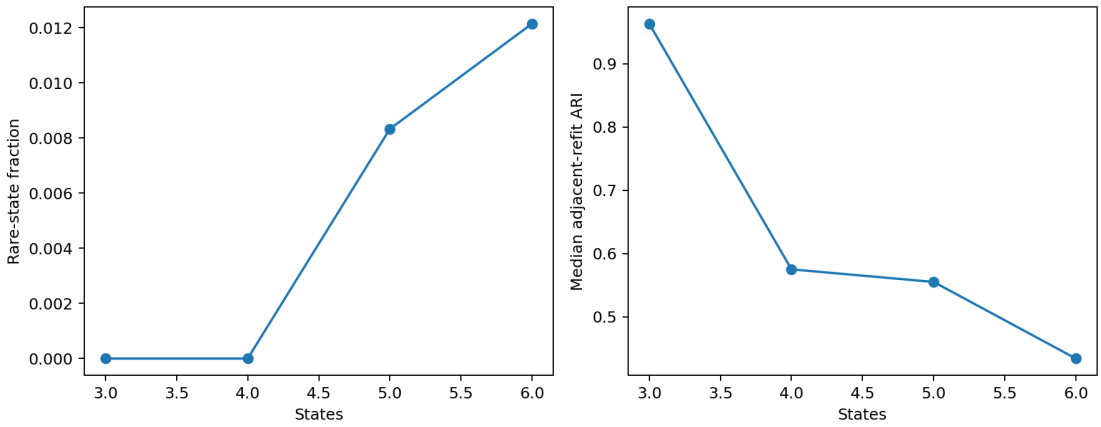


Figure 2: Rare occupancy and adjacent-refit stability under the fixed diagnostic rules.

Table 4: Controlled HMM ablations and weighting specifications. NoProbability removes P but retains regime multipliers; Entropy removes multipliers but retains P; NoHMM removes both.

Model	Dates	RMSE	95% block CI	MAE	AUC
Adaptive	1810	0.1053	[0.0557, 0.1555]	0.0569	0.6583
Entropy	1810	0.1048	[0.0554, 0.1546]	0.0568	0.6656
EntropyDeficit	1810	0.1072	[0.0580, 0.1577]	0.0593	0.6421
Equal	1810	0.1046	[0.0554, 0.1542]	0.0568	0.6776
NoEVT	1810	0.1068	[0.0583, 0.1560]	0.0597	0.5919
NoGARCH	1810	0.1058	[0.0559, 0.1563]	0.0572	0.6468
NoHMM	1810	0.1041	[0.0547, 0.1540]	0.0564	0.6573
NoProbability	1810	0.1052	[0.0553, 0.1552]	0.0569	0.6527
PCA	1810	0.1053	[0.0553, 0.1555]	0.0570	0.6843
PCAFloor	1810	0.1049	[0.0551, 0.1551]	0.0568	0.6836

Table 5: Adaptive versus controlled references. Positive gain favors Adaptive. Bootstrap intervals are marginal; adjusted p-values refer to squared-loss DM tests.

Reference	RMSE gain	CI lower	CI upper	DM p	Family Holm	Global Holm
NoProbability	-0.0001	-0.0026	0.0020	0.9092	1.0000	1.0000
Entropy	-0.0005	-0.0011	0.0002	0.2199	1.0000	1.0000
NoHMM	-0.0012	-0.0038	0.0012	0.3410	1.0000	1.0000
Equal	-0.0008	-0.0015	0.0003	0.2169	1.0000	1.0000
PCA	-0.0000	-0.0025	0.0021	0.9677	1.0000	1.0000
NoEVT	0.0015	-0.0016	0.0057	0.3847	1.0000	1.0000
NoGARCH	0.0005	-0.0002	0.0012	0.1795	1.0000	1.0000

Table 6: Exploratory conditional-information regressions. V/T and P/stress are separate predictors; these rows are not weighted composite specifications.

Regressors	Dates	RMSE	95% block CI	MAE	AUC	Brier
ConditionalVT	1810	0.1052	[0.0600, 0.1536]	0.0594	0.5965	0.2080
ConditionalVTP	1810	0.1068	[0.0601, 0.1568]	0.0598	0.5972	0.2085
ConditionalVTPStress	1810	0.1068	[0.0614, 0.1558]	0.0593	0.5843	0.2084
ConditionalVTStress	1810	0.1061	[0.0611, 0.1549]	0.0589	0.5866	0.2079

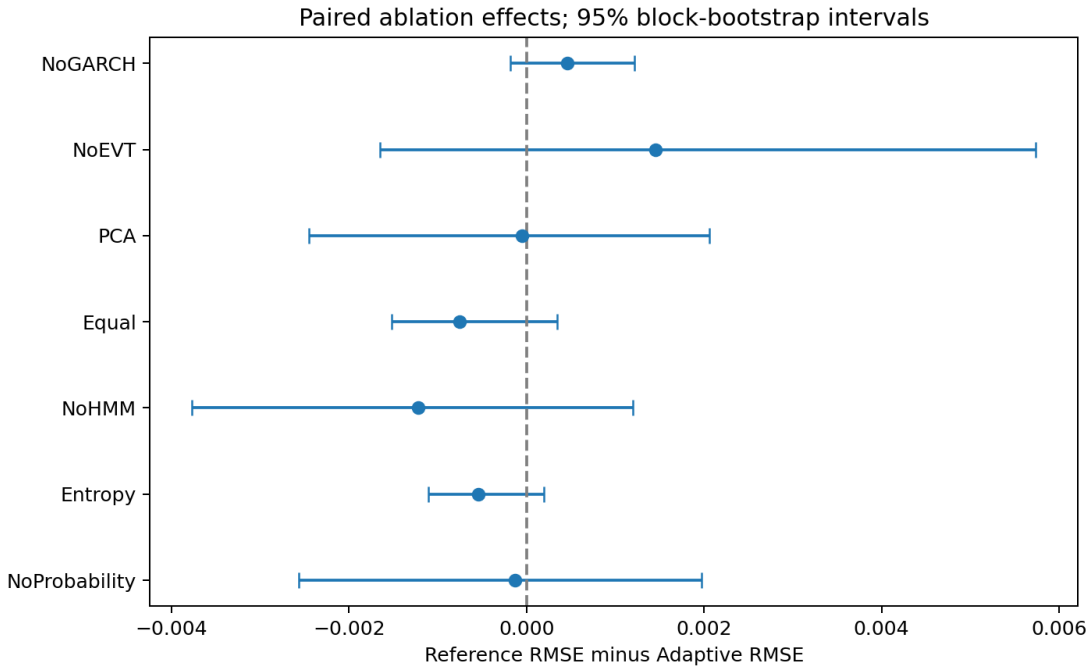


Figure 3: Paired RMSE improvements for Adaptive relative to controlled references. Positive values favor Adaptive.

6.4 EVT feasibility and sensitivity

Table 7: Empirical ES99 fallback frequency averaged across assets. An all-fallback cell is not a successful GPD robustness check.

Specification	Fallback fraction
w252_q90	0.0001
w252_q95	1.0000
w252_q97.5	1.0000
w378_q90	0.0000
w378_q95	1.0000
w378_q97.5	1.0000
w504_q90	0.0000
w504_q95	0.0000
w504_q97.5	1.0000

Five grid cells use empirical fallback for every asset/window estimate. At 95%/252 days, 95%/378 days and all three 97.5% configurations, the available number of excesses cannot meet the fixed minimum. These completed experiments reveal infeasible parametric specifications under the original rule; they are not missing experiments or evidence of successful GPD estimation. Identical all-fallback cells do not provide independent confirmations of threshold robustness.

Raw parameter and VaR/ES stability tables retain missing undefined estimates, fit failures and extreme finite ES. A stable normalized index can coexist with unstable raw tail estimates because rolling normalization and clipping compress differences. The score and forecast comparisons must therefore be interpreted together with the raw-tail diagnostics. The full data include one empirical fallback in the original 7,539 rolling fits; the new grid is assessed separately.

Table 8: Matched-date EVT-grid forecast evaluation. w denotes the POT window, q its threshold percentile; component normalization remains 252 days.

Tail specification	Dates	RMSE	95% block CI	MAE	AUC	Brier
RealizedVolBaseline	1570	0.1297	[0.0750, 0.1774]	0.0722	0.4195	0.2062
TrainMean	1570	0.1179	[0.0646, 0.1724]	0.0711	0.4195	0.2062
w252_q90	1570	0.1125	[0.0562, 0.1682]	0.0597	0.7015	0.1766
w252_q95	1570	0.1140	[0.0575, 0.1699]	0.0607	0.6855	0.1815
w252_q97.5	1570	0.1140	[0.0575, 0.1699]	0.0607	0.6855	0.1815
w378_q90	1570	0.1126	[0.0601, 0.1655]	0.0629	0.6617	0.1812
w378_q95	1570	0.1128	[0.0603, 0.1658]	0.0627	0.6609	0.1808
w378_q97.5	1570	0.1128	[0.0603, 0.1658]	0.0627	0.6609	0.1808
w504_q90	1570	0.1147	[0.0649, 0.1665]	0.0665	0.5227	0.2006
w504_q95	1570	0.1155	[0.0659, 0.1673]	0.0675	0.5140	0.2034
w504_q97.5	1570	0.1155	[0.0653, 0.1676]	0.0667	0.5308	0.2019

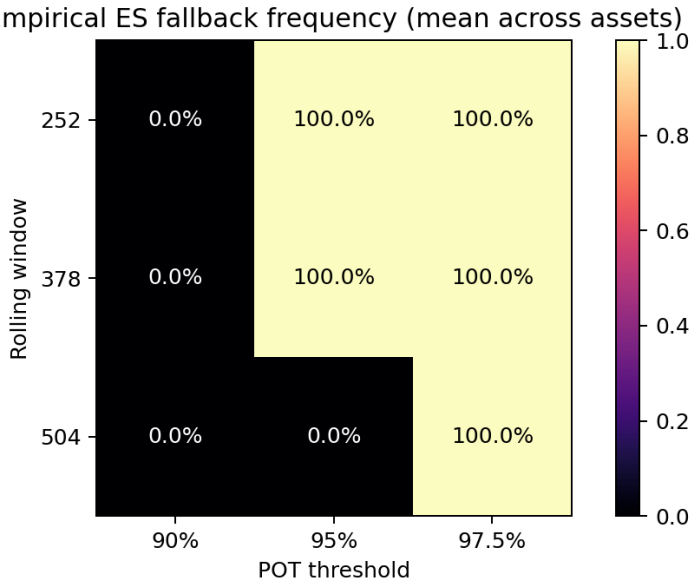


Figure 4: Fraction of operative tail estimates using empirical ES99 fallback.

6.5 Crisis prediction and descriptive context

Table 9: Future 30-day crisis prediction, fixed probability cutoff 0.5. These are not same-day detection scores.

Model	Precision	Recall	F1	AUC	95% AUC CI	Brier
Adaptive	0.2653	0.0835	0.1270	0.6583	[0.5257, 0.7803]	0.1869
Entropy	0.2313	0.0792	0.1180	0.6656	[0.5367, 0.7814]	0.1858
EntropyDeficit	0.3295	0.1242	0.1804	0.6421	[0.5122, 0.7613]	0.1941
Equal	0.4111	0.1585	0.2287	0.6776	[0.5421, 0.7977]	0.1842
NoEVT	0.0672	0.0171	0.0273	0.5919	[0.4691, 0.7111]	0.1982
NoGARCH	0.1534	0.0535	0.0794	0.6468	[0.5201, 0.7708]	0.1906
NoHMM	0.5322	0.1949	0.2853	0.6573	[0.5126, 0.7911]	0.1847
NoProbability	0.4903	0.1627	0.2444	0.6527	[0.5118, 0.7869]	0.1869
PCA	0.5181	0.1842	0.2717	0.6843	[0.5575, 0.8059]	0.1799
PCAFloor	0.5298	0.1906	0.2803	0.6836	[0.5560, 0.8024]	0.1800

The earlier after-close same-day detection F1 of 0.300 at an expanding 90th-percentile index threshold remains a different task from Table 9. A contemporaneous signal observes the day's return; it cannot demonstrate prevention or advance prediction of that loss. Positive future-event labels span long runs, so many positive daily labels do not imply many independent crisis episodes. The original five eligible early-warning events and conditional median crossing lead of 7.5 trading days remain exploratory, not newly validated population lead-time estimates.

The original descriptive wavelet, stationarity, structural-break and cointegration analyses are retained as context. Full-sample LOESS/wavelet transforms are not predictive inputs. The historical external FSI comparison uses revised histories and assumed release lags; its lower augmented RMSE is not treated as a vintage-correct benchmark advantage. None of these ancillary results resolves the limited breadth of two overlapping technology stocks and their benchmark ETF.

7 Discussion and limitations

The results distinguish construction from usefulness. A-CMSI gives an explicit multidimensional after-close stress summary, but a model assembled from economically plausible components need not minimize forecasting loss. Direct entropy weights reward sustained activity; deficit entropy rewards concentration. Regime factors impose analyst judgments about when tail severity matters more. Changing these choices can redistribute the same correlated information rather than add new signal.

The sixth state is evaluated on training likelihood, future emission density, regime diagnostics and index forecasts separately. A lower BIC is a penalized training criterion, not a guarantee of stable or useful states. Conversely, an interpretable five-state taxonomy is not statistical evidence of superiority. Any eventual production choice should be checked on a genuinely new period, not chosen from the most favorable retrospective metric in the present tables.

Tail inference is especially sensitive to feasibility. A 252-day 90th-percentile fit contains only roughly 25 excesses; moving to a higher quantile can exhaust the sample rather than improve tail specificity.

Longer windows add observations but assume greater temporal stability. The operational fallback is useful for keeping a signal defined, yet changes the estimator's statistical meaning. All-fallback robustness cells should be described as empirical-tail alternatives.

Several remaining limitations constrain the claims. Adjusted closes and published stress series are revised observations rather than verified historical vintages. The assets have overlapping economic exposures and cover neither credit nor funding markets. Crisis episodes are few. Forecast targets overlap and may remain dependent beyond the tested 30/60-day block choices. Bootstrap/HAC intervals are approximate under financial nonstationarity. HMM labels and transition targets are estimated constructs. A small number of selected fits have negative final likelihood increments; these are disclosed. The original log-volatility calibration is not an unbiased conditional-mean forecast on the original scale. Weight multipliers, clipping, state ranks and exceedance rules are design assumptions. Most importantly, the evaluation period has already been used in earlier research and cannot support a claim of independent external validation.

17 numerical and timing tests and artifact-level checks support implementation integrity, not economic truth of the models. A prefix test cannot establish that revised source prices were available historically. Better out-of-sample results on one loss cannot establish performance for all crisis-detection tasks. Non-rejection of equal loss cannot establish model equivalence. These distinctions are retained in the interpretation rather than hidden in supplementary material.

8 Conclusion

This revision reproduces the original A-CMSI experiment and executes the previously untested state-count, controlled-ablation, weighting and EVT-grid alternatives. All comparisons preserve chronological model fitting, mature-label calibration and fixed hyperparameters. The contribution is a transparent and reproducible stress-measurement framework with an explicit robustness audit, not proof of uniformly superior adaptive forecasting. Adaptive weighting has no demonstrated predictive advantage over simpler weighting baselines in this sample; failure to reject equal loss is not equivalence. HMM6 has the smaller point-estimate volatility RMSE. The five-versus-six comparison does not establish a family-adjusted volatility advantage; the RMSE difference remains uncertain. This is not proof of equivalence. Training BIC alone does not establish out-of-sample overfitting. Further work should use a new evaluation period, broader non-overlapping exposures, verified point-in-time observations and independently specified tail/state diagnostics before operational or strong systemic-risk claims are made.

Data and code availability

This paper revises the research archived under 10.5281/zenodo.23682477. The revised numerical results derive from the executed robustness experiments. Code is available at `https://github.com/ayushanand1103/A-CMSI-research`; audit commands, locks, data hashes, configuration and reports are packaged separately from original outputs. Third-party input data remain subject to their original terms.

Declarations

Ayush Anand is the sole author. This research was conducted independently and received no external funding. The author declares no competing interests. Generative-AI assistance was used for code, research checking, experiment implementation, manuscript drafting and formatting. The human author is responsible for verifying results, references, disclosures, licensing and final submission decisions. No AI system is listed as an author.

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